

Integral Calculus (2022–2026)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

26

TOTAL MARKS

42 M

TEST DURATION

78 Min

PAPER PATTERN

MCQ • MSQ • NAT
(2022–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	11	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	1	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	14	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q55 • ID: JAM_2026_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55	The volume of the tetrahedron bounded by the planes $x = 1, y = 2, z = 3$ and $12x + 8y + 6z = 70$ is _____ (rounded off to one decimal place).

Question 2 (JAM 2026 • Q53 • ID: JAM_2026_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53	The double integral of $f(x, y) = x$ over the triangular region with vertices at $\left(-\frac{1}{2}, \frac{1}{2}\right), (1, 2)$ and $(1, -1)$ is _____ (rounded off to one decimal place).
------	--

Question 3 (JAM 2026 • Q22 • ID: JAM_2026_Q22)

MCQ

+2 M

Neg: -0.67

Q.22	Let S be the solid of intersection of two solid spheres S_1, S_2 given by $S_1: x^2 + y^2 + (z - 1)^2 \leq 4 \quad \text{and} \quad S_2: x^2 + y^2 + (z + 1)^2 \leq 4.$ Which ONE of the following iterated integrals expresses the volume of S?
(A)	$8 \int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} (\sqrt{4-x^2-y^2} - 1) dy dx$
(B)	$8 \int_0^2 \int_0^{\sqrt{4-x^2}} (\sqrt{4-x^2-y^2} - 1) dy dx$
(C)	$2 \int_0^2 \int_0^{\sqrt{4-x^2}} (\sqrt{4-x^2-y^2} - 1) dy dx$
(D)	$2 \int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} (\sqrt{4-x^2-y^2} - 1) dy dx$

Question 4 (JAM 2026 • Q20 • ID: JAM_2026_Q20)

MCQ

+2 M

Neg: -0.67

Q.20	The semi-circle $y = \sqrt{6x - 5 - x^2}$ is rotated clockwise by the angle $\frac{\pi}{2}$ in the plane about the origin. What is the area of the planar region traced out by the semi-circle?
(A)	6π
(B)	4π
(C)	2π
(D)	π

Question 5 (JAM 2026 • Q10 • ID: JAM_2026_Q10)

MCQ

+1 M

Neg: -0.33

Q.10	Let T be the triangular region in the plane with vertices at $(0,0)$, $(1,0)$ and $(1,1)$. Let f be a function defined from T to $\mathbb{R}$ given in polar coordinates. Then the double integral of f over T is equal to _____.
(A)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr + \int_1^{\sqrt{2}} \left(\int_0^{\cos^{-1}(1/r)} f \, d\theta \right) r \, dr$
(B)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr - \int_1^{\sqrt{2}} \left(\int_0^{\cos^{-1}(1/r)} f \, d\theta \right) r \, dr$
(C)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr + \int_1^{\sqrt{2}} \left(\int_0^{\sin^{-1}(1/r)} f \, d\theta \right) r \, dr$
(D)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr - \int_1^{\sqrt{2}} \left(\int_0^{\sin^{-1}(1/r)} f \, d\theta \right) r \, dr$

Question 6 (JAM 2026 • Q5 • ID: JAM_2026_Q5)

MCQ

+1 M

Neg: -0.33

Q.5

Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function. Then

$$\int_0^2 \int_0^x \int_0^y f(x, y, z) dz dy dx = \text{_____}.$$

(A)

$$\int_0^2 \int_0^y \int_0^2 f(x, y, z) dx dz dy$$

(B)

$$\int_0^2 \int_0^x \int_z^2 f(x, y, z) dy dz dx$$

(C)

$$\int_0^2 \int_y^2 \int_0^x f(x, y, z) dz dx dy$$

(D)

$$\int_0^2 \int_z^2 \int_0^x f(x, y, z) dy dx dz$$

Question 7 (JAM 2025 • Q57 • ID: JAM_2025_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57

Let Ω be the solid bounded by the planes $z = 0$, $y = 0$, $x = \frac{1}{2}$, $2y = x$ and $2x + y + z = 4$.

If V is the volume of Ω , then the value of $64V$ is equal to _____
(rounded off to two decimal places).

Question 8 (JAM 2025 • Q48 • ID: JAM_2025_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let T denote the triangle in the xy plane bounded by the x axis and the lines $y = x$ and $x = 1$. The value of the double integral (over T)

$$\iint_T (5 - y) dx dy$$

is equal to _____ (rounded off to two decimal places).

Question 9 (JAM 2025 • Q45 • ID: JAM_2025_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45 Let S be the surface area of the portion of the plane $z = x + y + 3$, which lies inside the cylinder $x^2 + y^2 = 1$.

Then, the value of $\left(\frac{S}{\pi}\right)^2$ is equal to _____ (rounded off to two decimal places).

Question 10 (JAM 2025 • Q28 • ID: JAM_2025_Q28)

MCQ

+2 M

Neg: -0.67

Q.28 Let Ω be the bounded region in $\mathbb{R}^3$ lying in the first octant ($x \geq 0$, $y \geq 0$, $z \geq 0$), and bounded by the surfaces $z = x^2 + y^2$, $z = 4$, $x = 0$ and $y = 0$.

Then, the volume of Ω is equal to

- (A) π
- (B) 2π
- (C) 3π
- (D) 4π

Question 11 (JAM 2025 • Q10 • ID: JAM_2025_Q10)

MCQ

+1 M

Neg: -0.33

Q.10 The value of

$$\int_0^1 \left(\int_{\sqrt{y}}^1 3e^{x^3} dx \right) dy$$

is equal to

- (A) $e - 1$
- (B) $\frac{e-1}{2}$
- (C) $\sqrt{e} - 1$
- (D) $\frac{\sqrt{e}-1}{2}$

Question 12 (JAM 2024 • Q57 • ID: JAM_2024_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57 Let $c > 0$ be such that

$$\int_0^c e^{s^2} ds = 3$$

Then, the value of

$$\int_0^c \left(\int_x^c e^{x^2+y^2} dy \right) dx$$

equals _____ (rounded off to one decimal place).

Question 13 (JAM 2024 • Q54 • ID: JAM_2024_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54 Let

$$S = \{(x, y, z) \in \mathbb{R}^3: x^2 + y^2 + z^2 < 1\}.$$

Then, the value of

$$\frac{1}{\pi} \iiint_S \left((x - 2y + z)^2 + (2x - y - z)^2 + (x - y + 2z)^2 \right) dx dy dz$$

equals _____ (rounded off to two decimal places).

Question 14 (JAM 2024 • Q53 • ID: JAM_2024_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53 Let T be the planar region enclosed by the square with vertices at the points $(0,1)$, $(1,0)$, $(0,-1)$ and $(-1,0)$. Then, the value of

$$\iint_T \left(\cos(\pi(x-y)) - \cos(\pi(x+y)) \right)^2 dx dy$$

equals _____ (rounded off to two decimal places).

Question 15 (JAM 2024 • Q48 • ID: JAM_2024_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48 Let

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 4, (x-1)^2 + y^2 \leq 1, z \geq 0\}.$$

Then, the surface area of S equals _____ (rounded off to two decimal places).

Question 16 (JAM 2024 • Q41 • ID: JAM_2024_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41 The area of the region

$$R = \left\{ (x, y) \in \mathbb{R}^2 : 0 \leq x \leq 1, 0 \leq y \leq 1 \text{ and } \frac{1}{4} \leq xy \leq \frac{1}{2} \right\}$$

is _____ (rounded off to two decimal places).

Question 17 (JAM 2024 • Q27 • ID: JAM_2024_Q27)

MCQ

+2 M

Neg: -0.67

Q.27 For $a > b > 0$, consider

$$D = \left\{ (x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq a^2 \text{ and } x^2 + y^2 \geq b^2 \right\}.$$

Then, the surface area of the boundary of the solid D is

- (A) $4\pi(a+b)\sqrt{a^2-b^2}$
- (B) $4\pi(a^2-b\sqrt{a^2-b^2})$
- (C) $4\pi(a-b)\sqrt{a^2-b^2}$
- (D) $4\pi(a^2+b\sqrt{a^2-b^2})$

Question 18 (JAM 2023 • Q55 • ID: JAM_2023_Q55)

NAT

+2 M

Neg: Nil (0)

Q. 55 Let S be the triangular region whose vertices are $(0, 0)$, $(0, \frac{\pi}{2})$, and $(\frac{\pi}{2}, 0)$. The value of $\iint_S \sin(x) \cos(y) dx dy$ is equal to _____.
(rounded off to two decimal places)

Question 19 (JAM 2023 • Q48 • ID: JAM_2023_Q48)

NAT

+1 M

Neg: Nil (0)

Q. 48 Let V be the volume of the region $S \subseteq \mathbb{R}^3$ defined by

$$S = \{(x, y, z) \in \mathbb{R}^3 : xy \leq z \leq 4, 0 \leq x^2 + y^2 \leq 1\}$$

Then $\frac{V}{\pi}$ is equal to _____. (rounded off to two decimal places)

Question 20 (JAM 2023 • Q31 • ID: JAM_2023_Q31)

MSQ

+2 M

Neg: Nil (0)

Q. 31 For each $t \in (0, 1)$, the surface P_t in $\mathbb{R}^3$ is defined by

$$P_t = \{(x, y, z) : (x^2 + y^2)z = 1, t^2 \leq x^2 + y^2 \leq 1\}.$$

Let $a_t \in \mathbb{R}$ be the surface area of P_t . Then

(A) $a_t = \iint_{t^2 \leq x^2 + y^2 \leq 1} \sqrt{1 + \frac{4x^2}{(x^2 + y^2)^4} + \frac{4y^2}{(x^2 + y^2)^4}} dx dy$

(B) $a_t = \iint_{t^2 \leq x^2 + y^2 \leq 1} \sqrt{1 + \frac{4x^2}{(x^2 + y^2)^2} + \frac{4y^2}{(x^2 + y^2)^2}} dx dy$

(C) the limit $\lim_{t \rightarrow 0^+} a_t$ does NOT exist

(D) the limit $\lim_{t \rightarrow 0^+} a_t$ exists

Question 21 (JAM 2023 • Q26 • ID: JAM_2023_Q26)

MCQ

+2 M

Neg: -0.67

Q. 26 Let

$$f(x, y) = \iint_{(u-x)^2 + (v-y)^2 \leq 1} e^{-\sqrt{(u-x)^2 + (v-y)^2}} du dv.$$

Then $\lim_{n \rightarrow \infty} f(n, n^2)$ is

- (A) non-existent
- (B) 0
- (C) $\pi(1 - e^{-1})$
- (D) $2\pi(1 - 2e^{-1})$

Question 22 (JAM 2023 • Q18 • ID: JAM_2023_Q18)

MCQ

+2 M

Neg: -0.67

Q. 18 The area of the curved surface

$$S = \{(x, y, z) \in \mathbb{R}^3 : z^2 = (x-1)^2 + (y-2)^2\}$$

lying between the planes $z = 2$ and $z = 3$ is

- (A) $4\pi\sqrt{2}$
- (B) $5\pi\sqrt{2}$
- (C) 9π
- (D) $9\pi\sqrt{2}$

Question 23 (JAM 2023 • Q7 • ID: JAM_2023_Q7)

MCQ

+1 M

Neg: -0.33

Q. 7 The value of

$$\int_0^1 \int_0^{1-x} \cos(x^3 + y^2) dy dx - \int_0^1 \int_0^{1-y} \cos(x^3 + y^2) dx dy$$

is

- (A) 0
- (B) $\frac{\cos(1)}{2}$
- (C) $\frac{\sin(1)}{2}$
- (D) $\cos\left(\frac{1}{2}\right) - \sin\left(\frac{1}{2}\right)$

Question 24 (JAM 2022 • Q52 • ID: JAM_2022_Q52)

NAT

+2 M

Neg: Nil (0)

Q.52 If G is the region in $\mathbb{R}^2$ given by

$$G = \left\{ (x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1, \frac{x}{\sqrt{3}} < y < \sqrt{3}x, x > 0, y > 0 \right\}$$

then the value of

$$\frac{200}{\pi} \iint_G x^2 dx dy$$

is equal to _____. (Rounded off to two decimal places)

Question 25 (JAM 2022 • Q48 • ID: JAM_2022_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48

Consider the region

$$G = \{(x, y, z) \in \mathbb{R}^3 : 0 < z < x^2 - y^2, x^2 + y^2 < 1\}.$$

Then the volume of G is equal to _____. (Rounded off to two decimal places)

Question 26 (JAM 2022 • Q11 • ID: JAM_2022_Q11)

MCQ

+2 M

Neg: -0.67

Q.11

Consider the open rectangle $G = \{(s, t) \in \mathbb{R}^2 : 0 < s < 1 \text{ and } 0 < t < 1\}$ and the map $T: G \rightarrow \mathbb{R}^2$ given by

$$T(s, t) = \left(\frac{\pi s(1-t)}{2}, \frac{\pi(1-s)}{2} \right) \quad \text{for } (s, t) \in G.$$

Then the area of the image $T(G)$ of the map T is equal to

(A)

$$\frac{\pi}{4}$$

(B)

$$\frac{\pi^2}{4}$$

(C)

$$\frac{\pi^2}{8}$$

(D)

$$1$$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Integral Calculus (2022–2026) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	4.0 to 4.0
2	NAT	+2	1.1 to 1.3
3	MCQ	+2	A
4	MCQ	+2	A
5	MCQ	+1	MTA
6	MCQ	+1	A
7	NAT	+2	MTA
8	NAT	+1	2.31 to 2.35
9	NAT	+1	2.98 to 3.02
10	MCQ	+2	B
11	MCQ	+1	A
12	NAT	+2	4.4 to 4.6
13	NAT	+2	4.70 to 4.90

Q. No.	Type	Marks	Official Key
14	NAT	+2	1.95 to 2.05
15	NAT	+1	4.50 to 4.60
16	NAT	+1	0.24 to 0.26
17	MCQ	+2	A
18	NAT	+2	0.49 to 0.51
19	NAT	+1	3.99 to 4.01
20	MSQ	+2	A, C
21	MCQ	+2	D
22	MCQ	+2	B
23	MCQ	+1	A
24	NAT	+2	4.15 to 4.17
25	NAT	+1	0.49 to 0.51
26	MCQ	+2	C

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.